

D8 – Annex Q

Cleaned Scientific Source Basis

Version 3.0 – Continuous Numbering, Scientific Papers Only

1. Status and Function

D8 belongs to the source basis of the set.

It contains exclusively scientific papers that are relevant to the theoretical, operational, critical, or extrapolative structure of D1–D7.

Not included are:

- books,
- unpublished manuscripts,
- uncertain working versions,
- sources that cannot be unambiguously bibliographically reconstructed,
- preprints, unless they are expressly to be listed in a separate early-indicator list.

D8 does not replace any argumentation in D1–D7.

It documents the scientific connection points of the framework.

2. Source List Q1–Q14

Q1

Wong, M. L., Cleland, C. E., Arend, D. Jr., Bartlett, S., Cleaves II, H. J., Demarest, H., Prabhu, A., Lunine, J. I., & Hazen, R. M. (2023).

On the roles of function and selection in evolving systems.

Proceedings of the National Academy of Sciences, 120(43), e2310223120.

<https://doi.org/10.1073/pnas.2310223120>

Function in the Set: Foundation for functional information, selection, evolving systems, and the D1 axiomatics on functional structural development. (pnas.org)

Q2

Wong, M. L., Demarest, H., Bartlett, S., Cleland, C. E., Cleaves II, H. J., Prabhu, A., & Hazen, R. M. (2025).

Selective funnelling and state-space expansion: A conceptual framework for novelty generation and origin events.

Interface Focus, 15(6), 20250009.

<https://doi.org/10.1098/rsfs.2025.0009>

Function in the Set: Foundation for the D1 double movement of Selective Funnelling and State-Space Expansion. (royalsocietypublishing.org)

Q3

Hoel, E. P., Albantakis, L., & Tononi, G. (2013).

Quantifying causal emergence shows that macro can beat micro.

Proceedings of the National Academy of Sciences, 110(49), 19790–19795.

<https://doi.org/10.1073/pnas.1314922110>

Function in the Set: Source for causal emergence, multi-scale description, and the possibility that macro levels can be more relevant for explanation or control than micro levels. (pnas.org)

Q4

Hoel, E. P. (2017).

When the map is better than the territory.

Entropy, 19(5), 188.

<https://doi.org/10.3390/e19050188>

Function in the Set: Source for macroscopic modeling, causal condensation, and the D3 question of when higher description levels are operationally more viable. (MDPI)

Q5

Tononi, G. (2004).

An information integration theory of consciousness.

BMC Neuroscience, 5, 42.

<https://doi.org/10.1186/1471-2202-5-42>

Function in the Set: Source for information, integration, and consciousness reference in D5/A1 as well as for the general question of multi-scale integration. (PubMed)

Q6

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017).

Attention Is All You Need.

Advances in Neural Information Processing Systems, 30.

Function in the Set: Source for Transformer architecture, attention mechanisms, and the technical foundation of modern AI systems in the context of D4/D4a and D7. (NeurIPS Papers)

Q7

Bastos, A. M., Vezoli, J., & Fries, P. (2015).

Communication through coherence with inter-areal delays.

Current Opinion in Neurobiology, 31, 173–180.

<https://doi.org/10.1016/j.conb.2014.11.001>

Function in the Set: Source for resonance coupling, non-zero-phase communication, inter-areal delays, feedforward/feedback differentiation, and D3/D3a examination variables on coupling quality. (PubMed)

Q8

Fries, P. (2015).

Rhythms for cognition: Communication through coherence.

Neuron, 88(1), 220–235.

<https://doi.org/10.1016/j.neuron.2015.09.034>

Function in the Set: Source for rhythmic communication, Gamma/Beta/Alpha/Theta modes, selective attention, cognitive updating, and D4a resonance hygiene. (PubMed)

Q9

Palacios, E. R., Razi, A., Parr, T., Kirchhoff, M., & Friston, K. (2020).

On Markov blankets and hierarchical self-organisation.

Journal of Theoretical Biology, 486, 110089.

<https://doi.org/10.1016/j.jtbi.2019.110089>

Function in the Set: Source for boundary formation, Markov Blankets, hierarchical self-organization, mediated inner-outer coupling, and D4/D4a interface logic. (PubMed)

Q10

Rosas, F., Mediano, P. A. M., Ugarte, M., & Jensen, H. J. (2018).

An information-theoretic approach to self-organisation: Emergence of complex interdependencies in coupled dynamical systems.

Entropy, 20(10), 793.

<https://doi.org/10.3390/e20100793>

Function in the Set: Source for Binding Information, interdependency formation, redundancy/synergy, and the distinction of genuine self-organization from mere entropy reduction. (College of Engineering)

Q11

Cui, H., & Yasseri, T. (2024).

AI-enhanced collective intelligence.

Patterns, 5(11), 101074.

<https://doi.org/10.1016/j.patter.2024.101074>

Function in the Set: Main source for D4/D4a: Human–AI Collective Intelligence, multilayer networks, AI roles, Trust, Explainability, Bias, Crowd-Retention, and human–AI interface risks. (Cell)

Q12

Hancock, F., Rosas, F. E., Luppi, A. I., Zhang, M., Mediano, P. A. M., Cabral, J., Deco, G., Kringelbach, M. L., Breakspear, M., Kelso, J. A. S., & Turkheimer, F. E. (2025).

Metastability demystified — the foundational past, the pragmatic present and the promising future.

Nature Reviews Neuroscience, 26(2), 82–100.

<https://doi.org/10.1038/s41583-024-00883-1>

Function in the Set: Source for metastability, integration/segregation, transient states, methodological limits of signatures, and D3a HOLD as operational proximity to metastable lingering. (PubMed)

Q13

Axenie, C., López-Corona, O., Makridis, M. A., Akbarzadeh, M., Saveriano, M., Stancu, A., & West, J. (2024).

Antifragility in complex dynamical systems.

npj Complexity, 1, 12.

<https://doi.org/10.1038/s44260-024-00014-y>

Function in the Set: Source for antifragility, learning from disturbance, robust adaptation, and D5/A5. (Nature)

Q14

Rovenskaya, E., Ivanov, A., Hathiari, S., Kotova, D., Scharler, U. M., & Boza, G. (2025).

An ecological perspective to master the complexities of the digital economy.

npj Complexity, 2, 16.

<https://doi.org/10.1038/s44260-025-00036-0>

Function in the Set: Source for complex adaptive systems, digital platform ecosystems, 5M-

Framework, D5/A3, and institutional-technical multi-scale analysis. (Nature)

3. Layer Assignment

D1 – Core Layer

Primarily relevant: Q1, Q2, Q3, Q4, Q7, Q8, Q9, Q10, Q12.

Function: Scientific connectability for functional evolution, multi-scale causality, resonance coupling, boundary formation, interdependency, and metastable complexity.

D2 – Admissibility Layer

Primarily relevant: Q9, Q10, Q11, Q12.

Function: No source sets norms; however, they clarify why complexity formation, resonance, self-organization, or collective intelligence are not automatically admissible.

D3 – Operational Layer

Primarily relevant: Q3, Q4, Q7, Q8, Q9, Q10, Q12.

Function: Proxy fields for condensation, reality coupling, path examination, boundary integrity, interdependency profile, and metastability range.

D3a – Operational Deep Layer

Primarily relevant: Q7, Q8, Q9, Q10, Q11, Q12.

Function: HOLD, reversible learning loops, resonance quality, boundary integrity, interdependency profile, human–AI coupling, and metastable steering under uncertainty.

D4 – Human & Interface

Primarily relevant: Q6, Q8, Q9, Q11, Q12.

Function: Human–system coupling, attention, interface, AI roles, Trust, Explainability, and preservation of real judgment agency.

D4a – Operational Coupling

Primarily relevant: Q7, Q8, Q9, Q10, Q11, Q12.

Function: Path-Risk-Scan, Pre-Mortem, AgencyScore, dissent space, resonance hygiene, minimal coupling dashboards, and protection against covert substitution.

D5 – Unfolding Layer

Primarily relevant: Q5, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14.

Function: Application fields: brain, consciousness, AI, platform economy, antifragility, collective intelligence, and digital ecosystems.

D6 – Critical Counter-Examination

Primarily relevant: Q7, Q8, Q9, Q10, Q11, Q12, Q13.

Function: Pseudo-resonance, boundary collapse, order illusion, metastability misgrasp, collective disempowerment, Proxy Break, and epistemic erosion.

D7 – Extrapolation Layer

Primarily relevant: Q1, Q2, Q3, Q4, Q6, Q7, Q8, Q9, Q10, Q11, Q12.

Function: Open connection spaces for physical, technical, AI-related, quantum-informational, and societal extrapolations.

4. Not in the Main Numbering Circle

The following sources should not appear in the D8 main numbering circle because D8 is limited here to scientific papers:

- Hazen & Wong, *Time's Second Arrow* — Book / not listed as a paper.
- Taleb, *Antifragile* — Book.
- Darwin, *On the Origin of Species* — Book.
- Korzybski, *Science and Sanity* — Book.
- Hubinger et al., *Risks from Learned Optimization* — relevant, but as preprint or non-peer-reviewed main source better placed in D8b.
- Greenblatt et al., *Alignment faking in large language models* — relevant, but as preprint / early indicator better placed in D8b.
- Koondijk, strategic behavior / Alignment Faking — only in D8b, as long as bibliographically and publication-wise not cleanly stabilized.
- Zhao et al., Quantum advantage massive classical data — as preprint / early indicator in D8b.